ML BASED WATER QUALITY PREDICTION FOR AQUACULTURE

MINOR PROJECT-1 REPORT

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BONAFIDE CERTIFICATE

Certified that this Minor Project-1 report entitled **"ML BASED WATER QUALITY PREDICTION FOR AQUACULTURE"** is the bonafide work of **"G.SRINIVASULU (22UEEA2022), M.PRAVEENREDDY (22UEEA2011) and Y.GANESH(22UEEA2013)"** who carried out the project work under my supervision.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

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ABSTRACT

Water quality plays a crucial role in sustainable aquaculture and agriculture. Poor water conditions can lead to serious consequences, including reduced crop yield and outbreaks of fish diseases, which in turn cause economic losses and environmental imbalances. This project proposes a machine learning (ML)-based solution to monitor and predict water quality parameters that are directly linked to fish health and agricultural productivity. Using a powerful ensemble learning method of the random forest algorithm, the key indicators of water quality such as pH, dissolved oxygen, temperature, ammonia, nitrates, turbidity and the presence of pathogens or toxins. Machine learning algorithms were trained using labeled datasets that classify water quality as either "Safe" or "Unsafe" based on known parameter combinations. Among all the algorithms like Support vector Machine, Naive Bayes, Logistic regression, etc, the Random Forest algorithm demonstrated the highest accuracy. A web application was developed using this trained model, allowing users—such as fish farmers, aquaculture technicians, or researchers—to upload water quality data, view predictions about fish health, water quality and receive recommendations or alerts based on the results.

This project proposes a machine learning approach for early detection of fish diseases by analyzing water quality with accuracy of 92 percentage and designing an interactive web page to analyse the water quality. Using a Random Forest algorithm trained on water quality data, the objective is to predict fish health by assessing toxins and viruses in water samples. This proactive method allows real-time monitoring and management of fish health, helping to prevent disease outbreaks and maintain healthy fish populations in aquaculture. Aquaculture plays a significant role in global food production, but fish diseases pose a major threat to the industry, leading to substantial economic losses and jeopardizing food security.

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CHAPTER 1

INTRODUCTION

Water quality influences every aspect of aquaculture from growth rates and feed efficiency to disease resistance and survival. Parameters such as pH, temperature, dissolved oxygen, ammonia, nitrites, nitrates, turbidity, and salinity must be carefully monitored and controlled. If these factors fall outside the suitable range, it can lead to stress, reduced growth, or even mass mortality in fish populations. Figure:1.1 shows the life cycle the fishes.

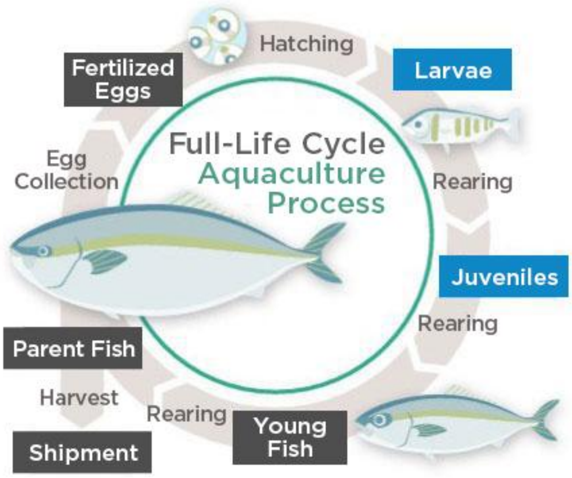

Figure 1.1: Life Cycle

1.1 ROLE OF MACHINE LEARNING IN WATER QUALITY PREDICTION

Aquaculture, the controlled cultivation of aquatic organisms such as fish, shrimp, and shellfish, is one of the fastest-growing sectors in global food production. With increasing demand for seafood and the depletion of natural fish stocks, aquaculture offers a sustainable solution to meet global protein needs. However, one of the most critical challenges in aquaculture is maintaining optimal water quality, as the health and productivity of aquatic organisms are directly linked to the physical, chemical, and biological conditions of their environment. Traditionally, water quality monitoring in aquaculture has been done manually or with basic automated systems. These methods are often labor-intensive, slow to respond, and not capable of providing early warnings. In many cases, intervention only occurs after visible signs of fish stress or disease, by which time the damage has already been done. Hence, there is a strong need for an intelligent, real-time prediction system that can proactively monitor and assess water conditions.

1.1.1 Advantages of ML-Based Water Quality Prediction

Machine Learning (ML)-based water quality prediction provides numerous advantages for aquaculture, enhancing efficiency, sustainability, and productivity. One of the primary benefits is real-time monitoring, where ML algorithms analyze sensor data continuously, allowing farmers to assess water conditions instantly. Traditional methods often require manual sampling and laboratory analysis, which can be time-consuming and prone to errors. ML automates this process, ensuring that water quality parameters such as temperature, pH, dissolved oxygen, and ammonia levels are monitored with precision. Another crucial advantage is early anomaly detection, which helps prevent sudden environmental changes that could harm aquatic life. ML models can identify patterns and trends in water quality data, predicting potential issues before they escalate. For example, if oxygen levels start to decline, ML systems can trigger aeration mechanisms automatically, preventing fish stress and mortality. This proactive approach minimizes losses and enhances the overall health of the aquatic ecosystem.

1.2 WATER QUALITY PARAMETERS

Water quality parameters are crucial factors in aquaculture, as they directly impact the health and growth of aquatic organisms. Proper monitoring and management of these parameters ensure a stable environment, reducing the risk of disease outbreaks and mortality. Water quality parameters play a crucial role in aquaculture, influencing the health, growth, and productivity of aquatic organisms. Monitoring and maintaining optimal water conditions ensures a stable environment, reducing the risk of disease outbreaks and mortality. One of the most important parameters is temperature, as it directly affects metabolic rates, oxygen solubility, and overall biological functions. Sudden fluctuations can lead to stress, poor growth, and even fatalities, making temperature regulation essential for successful aquaculture operations. pH levels are another critical factor, determining the acidity or alkalinity of water. Most aquatic species thrive in a neutral to slightly alkaline pH range between 6.5 and 8.5. Extreme pH variations can disrupt biological functions, reduce growth rates, and increase susceptibility to disease. Similarly, Dissolved Oxygen (DO) is fundamental for respiration in fish and other aquatic organisms. Low oxygen levels can lead to suffocation, stress, and reduced activity, making adequate aeration and water movement essential for maintaining proper oxygen levels. article

Table 1.1: Parameter Ranges for Best, Average, and Worst Case Scenarios

Parameter	**Best Case (2) Range**	**Average Case (1) Range**	**Worst Case (0) Range**
Temp (°C)	1.10 – 34.13	15.48 – 30.90	21.29 – 29.13
Turbidity (cm)	18.77 – 69.92	15.17 – 29.48	30.02 – 73.71
DO (mg/L)	2.34 – 7.51	5.27 – 7.68	3.95 – 4.91
BOD (mg/L)	1.25 – 14.54	2.19 – 5.37	1.01 – 1.99
CO_2 (mg/L)	2.99 – 14.93	1.21 – 9.75	5.66 – 7.85
pH	6.08 – 9.23	6.08 – 9.46	6.75 – 8.98
Alkalinity (mg/L)	26.27 – 168.73	33.24 – 196.93	25.67 – 95.16
Hardness (mg/L)	44.79 – 298.56	30.86 – 271.24	77.25 – 147.73
Calcium (mg/L)	9.86 – 184.13	13.34 – 161.85	28.08 – 94.24
Ammonia (mg/L)	0.00017 – 0.06013	0.02544 – 0.04552	0.00079 – 0.02466
Nitrite (mg/L)	0.00362 – 4.92	0.33 – 1.84	0.00043 – 0.01765
Phosphorus (mg/L)	0.01955 – 2.74	0.01245 – 2.83	0.19901 – 1.88
H_2S (mg/L)	0.00214 – 0.06013	0.00357 – 0.01808	0.01922 – 0.01993
Plankton (No./L)	78.60 – 5177.67	2091.39 – 5800.40	3006.29 – 4240.28

1.3 MACHINE LEARNING ALGORITHMS

Machine learning algorithms are essential tools that enable computers to learn from data and improve their performance over time without explicit programming. These algorithms fall into three main categories as shown in Figure 1.2 supervised learning, unsupervised learning, and reinforcement learning. In supervised learning, models are trained on labeled data, meaning they learn from examples with known outputs. Common algorithms in this category include decision trees, Support Vector Machines (SVM), and neural networks, which are widely used in tasks like image recognition and natural language processing. Unsupervised learning, on the other hand, deals with unlabeled data, where the model identifies patterns and structures without predefined labels. Clustering algorithms such as K-means help uncover hidden relationships in data, making them useful in fields like customer segmentation and anomaly detection. Reinforcement learning differs from both by allowing models to learn through interaction with an environment, receiving rewards or penalties based on their actions. This approach is particularly useful in robotics, gaming, and autonomous systems, where continuous learning and adaptation are essential.

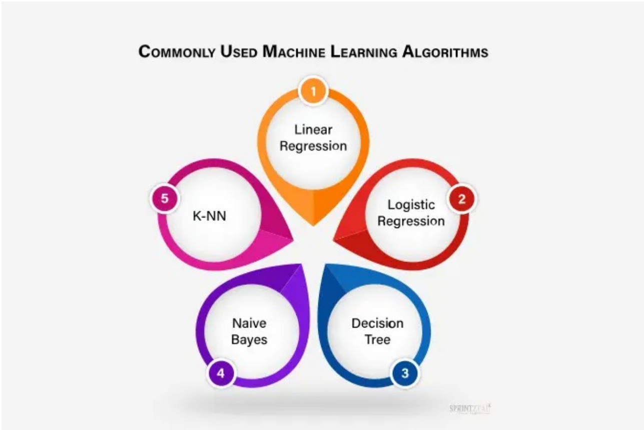

Figure 1.2: ML Algorithms

1.4 NEED FOR THE PROJECT

Machine learning-based water quality prediction for aquaculture is essential to ensuring a sustainable and efficient farming environment. Water quality plays a critical role in the health, growth, and survival of aquatic organisms, and any fluctuations in key parameters such as pH, dissolved oxygen, temperature, and ammonia levels can have significant consequences. Traditional water monitoring methods are often manual and time-consuming, making it difficult to respond to changes quickly. By leveraging machine learning techniques, aquaculture farmers can benefit from real-time monitoring and predictive insights that allow them to take proactive measures, reducing losses due to poor water conditions. Additionally, ML models enhance sustainability by optimizing resource usage, ensuring that interventions are only made when necessary.

1.5 OBJECTIVE OF THE PROJECT

To enhance the efficiency and sustainability of aquaculture practices by leveraging machine learning techniques. This project aims to monitor, analyze, and predict key water quality parameters such as pH, dissolved oxygen, temperature, turbidity, and ammonia levels to ensure optimal conditions for aquatic organisms. By integrating IoT sensors and ML algorithms, the system provides real-time insights and predictive alerts, enabling farmers to take proactive measures to prevent disease outbreaks and reduce mortality rates. Additionally, the project seeks to automate water quality assessment, minimizing manual intervention and improving overall productivity. The ultimate goal is to create a data-driven, intelligent system that supports sustainable aquaculture, optimizes resource utilization, and enhances yield while reducing environmental impact.

1.6 APPLICATIONS

Real-time Monitoring

Disease Prevention

Optimized Feeding Strategies

Automated Water Quality Management

1.7 SUMMARY

Machine learning enhances water quality monitoring in aquaculture, ensuring optimal conditions for fish health and growth. Traditional monitoring methods are manual and slow, whereas ML-based systems provide real-time insights, enabling proactive intervention. Key parameters like pH, dissolved oxygen, temperature, and ammonia are continuously analyzed, reducing stress and disease risks. Predictive models detect anomalies early, helping farmers optimize feeding and water management strategies. This technology supports sustainability by minimizing losses, improving productivity, and ensuring efficient resource utilization.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

Aquaculture, the cultivation of aquatic organisms under controlled conditions, is increasingly vital for meeting the global demand for seafood. A critical aspect of sustainable aquaculture is the continuous monitoring and prediction of water quality, which directly affects the health and productivity of aquatic species. Traditional water monitoring methods are often time-consuming, reactive rather than proactive, and resource-intensive. With the advancement of sensor technologies and artificial intelligence, machine learning (ML) models have emerged as effective tools for water quality prediction, enabling timely decision-making and optimizing aquaculture management practices.

2.2 LITERATURE SURVEY

Recent studies highlight various ML techniques applied to water quality prediction in aqua culture. One of the foundational contributions in this area is the comprehensive review by Li et al. (2020), which examined several machine learning models such as Support Vector Machines (SVM), Artificial Neural Networks (ANN), and Random Forest (RF). Their work emphasized the importance of key water quality parameters—pH, dissolved oxygen (DO), temperature, turbidity, and salinity—as indicators of aquaculture health. This review provided insights into the predictive performance of various models and underscored the need for real-time sensor data integration to enhance prediction accuracy. Raj et al. (2021) conducted an empirical study using Random Forest for predicting water quality in shrimp farms in India. They used real-time sensor data collected from local aquaculture ponds and applied RF to predict multiple parameters, including pH, temperature, and DO. The study compared RF with SVM and Decision Trees, ultimately finding that the Random Forest model delivered superior performance with an accuracy of around 95 In a more advanced approach, Nguyen et al. (2022) employed a Deep Neural Network (DNN) for predicting the Water Quality Index (WQI) in aquaculture systems. Their model incorporated over ten features, such as nitrate, ammonia, turbidity, and conductivity, drawn from both physical and chemical properties of water. The dataset also included temporal data sourced from IoT-based sensor systems. The study demonstrated that DNNs are effective in capturing complex nonlinear relation 6 ships among variables, outperforming classical models like linear regression or SVM in both training and testing phases. This research signaled a shift towards leveraging deep learning in aquaculture, especially where multidimensional datasets are involved. Expanding the scope beyond pure prediction, Yadav and Patel (2021) proposed a smart aquaculture system that integrated IoT sensors with machine learning algorithms for real-time water quality monitoring. The system used sensors to measure parameters such as pH, turbidity, ammonia levels, and temperature, and processed the data through K-Nearest Neighbors (KNN) and Gradient Boosting models. Their architecture also featured cloud-based services for remote monitoring and instant alerts, making it highly scalable for deployment in rural and remote aquaculture farms. This work is a prime example of how ML and IoT technologies can work in tandem to provide actionable insights and improve operational efficiency. Time-series prediction is another critical research direction, particularly useful for forecasting sudden changes in water quality that could harm aquatic life. Zhao et al. (2023) applied Long Short Term Memory (LSTM) neural networks to forecast water quality metrics using three years of hourly data. Their model focused on detecting early warning signs of harmful events, such as ammonia spikes or oxygen depletion, which are often precursors to mass mortality events in aquaculture systems. The LSTM model exhibited a low Root Mean Square Error (RMSE), indicating high accuracy and reliability in time-series forecasting. Their work illustrated the power of recurrent neural networks in anticipating dynamic water quality changes in aquaculture systems.Machine learning has emerged as a powerful tool for predicting water quality in aquaculture, enabling efficient monitoring and management of aquatic environments. Several studies have explored different ML techniques to enhance water quality prediction accuracy.

One notable study by Meghana S, Neha Rani Singh, Nethravathi A S, and Pallavi Lakshmi H L introduced a water quality prediction system that analyzes various water parameters to provide insights into maintaining suitable conditions for aquaculture. Their approach leverages sensor-based data collection and ML algorithms to predict fluctuations in water quality. Another research conducted by Rubén Baena-Navarro et al. integrates IoT sensors and ML algorithms, such as Random Forest, to predict water quality factors like temperature, dissolved oxygen, pH, and turbidity. Their study demonstrates high accuracy in predictions and real-time monitoring, ensuring optimal conditions for fish farming. Additionally, Zambrano AF and colleagues investigated machine learning models like Random Forests, Multivariate Linear Regression, and Artificial Neural Networks for predicting water conditions in fish farms. Their study focuses on scenarios where real-time monitoring devices are unavailable, making ML-based predictions crucial for manual water quality assessments. Table 2.1 shows the table presents of comparative literature survey analysis of AI-driven research in the field of nutrition and dietary assessment. It showcases various studies that explore personalized nutrition, AI-based food monitoring, and dietary recommendation systems.

Table 2.1: Literature Survey

S. NO	TITLE	PUBLISHED DETAILS	DESCRIPTION	INFERENCE
1	Personalized Diet Recommendation System Using Machine Learning	Author: D. Navya Narayana Kumari, T. Praveen Satya Published in: IJERT, 2024	Methodology: Used Nearest Neighbor model with cosine similarity. Materials: Streamlit for UI, FastAPI for backend. Limitations: Does not factor in disease levels like diabetes or real-time medical conditions.	The system successfully delivers personalized diet plans and nutritional recommendations to users, aiding in health management and weight control
2	Machine Learning in Nutrition Research	Authors: Daniel Kirk, Esther Kok, Michele Tufano Published in: IJERT, 2024	Methodology: Reviewed ML applications, applied ML on SPT data Materials: Streamlit Limitations: The performance of these models in distinct populations and with different data generation techniques is not guaranteed.	ML methods can significantly reduce the time and cost associated with traditional material tests by providing accurate predictions of material properties
3	Diet Recommendation System Using Machine Learning	Authors: Yuvraj Chibber, Dushyant Betala, Srividhya S. Published in: International Conference on Recent Trends in Data Science and its Applications, 2022	Methodology: Machine Learning algorithms, K-Means Algorithm. Materials: 90 food items, calories, fat, sugar, calcium, potassium. Limitations: Limited to 90 items, specific deficiencies, only on dietary recommendations.	The system effectively provides personalized diet plans based on user data, helping manage nutritional deficiencies and maintain a balanced diet.
4	AI-Based Nutritional Analysis	Author: R. Sharma, A. Gupta Published in: IEEE, 20242	Methodology: Deep Learning for food recognition. Materials: Large dataset of food images. Limitations: Accuracy varies with image quality.	AI-powered systems can automate dietary tracking and enhance food intake monitoring.

S. NO	TITLE	PUBLISHED DETAILS	DESCRIPTION	INFERENCE
5	ML-Based Food Recommendation System	Author: J. Lee, M. Kumar Published in: ACM, 2023	Methodology: Collaborative filtering and content-based recommendation. Materials: Nutrient database, user preferences. Limitations: Limited cultural dietary adaptation.	Enhances user experience by providing customized diet recommendations based on health goals.
6	AI in Personalized Dietary Recommendations	Author: T. Armand, A. Nfor Published in: MDPI, 2024	Methodology: Reviewed ML applications, applied ML on SPT data Materials: Streamlit Limitations: The performance of these models in distinct populations and with different data generation techniques is not guaranteed.	AI enhances personalized nutrition by tailoring diet plans based on individual health metrics.
7	AI's Role in Nutrition Research	Authors: Yuvraj Chibber, Dushyant Betala, Srividhya S. Published in: International Conference on Recent Trends in Data Science and its Applications, 2022	Methodology: Machine Learning algorithms, K-Means Algorithm. Materials: 90 food items, calories, fat, sugar, calcium, potassium. Limitations: Limited to 90 items, specific deficiencies, only on dietary recommendations.	The system effectively provides personalized diet plans based on user data, helping manage nutritional deficiencies and maintain a balanced diet.
8	AI-Based Nutritional Analysis	Author: A. Sosa-Holwerda, O. Park Published in: MDPI, 202	Methodology: Deep Learning for food recognition. Materials: Large dataset of food images. Limitations: Accuracy varies with image quality.	AI-powered systems can automate dietary tracking and enhance food intake monitoring.

2.3 SUMMARY

Machine learning plays a crucial role in aquaculture, enabling real-time monitoring and prediction of water quality parameters. Studies have demonstrated the effectiveness of various ML techniques, such as Random Forest, Deep Neural Networks, and LSTM, in accurately predicting changes in water conditions. IoT integration further enhances aquaculture management by providing automated alerts and scalable monitoring solutions. Research highlights the advantages of ML in reducing manual interventions, optimizing environmental stability, and improving fish health. As advancements continue, ML-driven approaches promise sustainable and efficient aquaculture practices.

CHAPTER 3

PROPOSED METHODOLOGY

Develop a machine learning-based approach aimed at early detection of fish diseases through the analysis of water quality.Train the Machine learning algorithms using a comprehensive water quality dataset to ensure accurate prediction of the health condition of fish.Utilize the trained algo rithms to provide an efficient and proactive method for detecting fish diseases, thereby aiding in the maintenance of aquatic ecosystem health. Figure 3.1 shows the operational block diagram as shown in below.

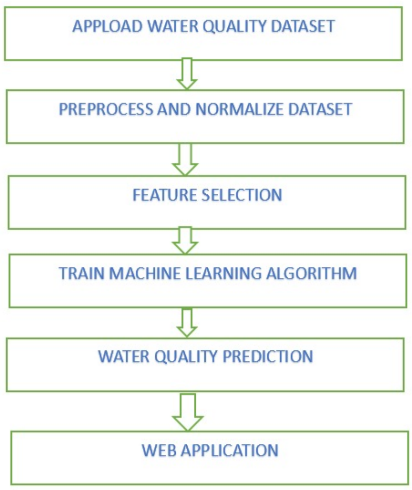

Figure 3.1: Operational Block Diagram

3.1 UPLOAD WATER QUALITY DATASET

The first step in the fish health prediction system is to upload the water quality dataset. This dataset serves as the foundational input for the entire process, providing essential environmental parameters that influence fish health.

3.2 PREPROCESS AND NORMALIZE DATASET

The second step in the system is to preprocess and normalize the dataset. Once the water quality data is uploaded, it often contains inconsistencies such as missing values, noise, or outliers that can negatively affect model performance. Preprocessing involves cleaning the dataset by handling missing data, correcting errors, and ensuring a consistent format across all records. This step may also include converting categorical values into numerical formats, removing duplicates, and standardizing units of measurement.

3.3 FEATURE SELECTION

The third step in the system is feature selection, a critical process that involves identifying the most relevant water quality parameters that significantly impact fish health. Not all features in a dataset contribute equally to the prediction outcome; some may be redundant or irrelevant, potentially leading to overfitting and reduced model accuracy.

3.4 MACHINE LEARNING ALGORITHM

The fourth step in the system is to train machine learning algorithms using the selected features from the previous step. During training, the algorithms analyze the input data and adjust their internal parameters to minimize prediction errors. Commonly used models in such applications include Random Forest, Support Vector Machine (SVM), Decision Trees.Each of these models has its strengths—some are better for handling large datasets, while others excel at recognizing complex patterns over time. The training process often involves splitting the dataset into training and validation sets to ensure the model can generalize well to unseen data.Table 3.1 shows the comparison of each algorithms based on accuracy.

Table 3.1: Comparison of Model Accuracies

Model	**Accuracy**
Logistic Regression	∼0.8
Decision Tree	∼0.95
Random Forest	∼0.95
SVM	∼0.9
Naive Bayes	∼0.85
KNN	∼0.85

Logistic Regression Classifier

Logistic regression is a statistical technique used primarily for binary classification problems, where the goal is to predict one of two possible outcomes based on input features. Despite its name, logistic regression is a classification algorithm, not a regression one. It models the probability that a given input belongs to a particular category.

Decision Tree Classifier

A decision tree classifier is a popular machine learning algorithm used for classification tasks, where the goal is to assign input data into predefined categories. It works by learning simple decision rules inferred from the data features. The model is structured like a tree, where each internal node represents a decision based on the value of a particular feature, each branch represents the outcome of that decision, and each leaf node represents a final class label.

Random Forest Classifier

Random Forest is a supervised machine learning algorithm that uses an ensemble of decision trees to make predictions. It's known for its accuracy, ability to handle both classification and regression tasks, and resistance to overfitting. The algorithm builds multiple decision trees, each trained on a random subset of the data, and then combines their predictions to arrive at a final result.

Support Vector Machine Classifier

A support vector machine (SVM) is a supervised machine learning algorithm that classifies data by finding an optimal line or hyperplane that maximizes the distance between each class in an N-dimensional space.

Naive Bayes classifier

Naive Bayes is a classification technique that is based on Bayes' Theorem with an assumption that all the features that predicts the target value are independent of each other.

K-Nearest Neighbors Classifier

The K-Nearest Neighbors (KNN) algorithm is a simple yet powerful machine learning technique used for both classification and regression tasks. It predicts the class or value of a new data point by considering its proximity to a certain number (K) of nearest neighbors in the training data.

RANDOM FOREST ALGORITHM

The Random Forest Classifier demonstrated the highest accuracy among the other algorithms utilized in the analysis. Additionally, it identified "pH" as the most influential feature in predicting disease presence based on water quality metrics. This underscores the significant role pH levels play in determining the likelihood of disease occurrence in aquatic environments. Maintaining optimal pH levels is crucial for preventing the proliferation of harmful pathogens and ensuring the overall health of aquatic ecosystems.Figure 3.2 shows the conclusion matrix for Random Forest Algorithm.

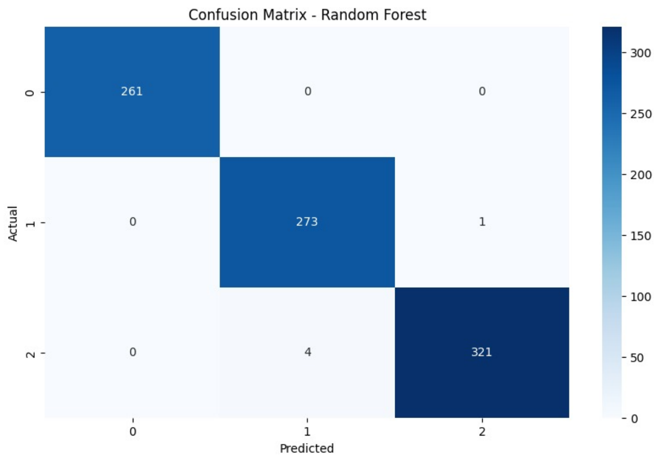

Figure 3.2: Random forest

Let $y_i \in \{0, 1\}$ be the true label, $\hat{y}_i \in \{0, 1\}$ the predicted label, and N the total number of samples. Define

$$\text{TP} = \sum_{i=1}^{N} \mathbf{1}\{y_i = 1 \wedge \hat{y}_i = 1\}, \quad \text{TN} = \sum_{i=1}^{N} \mathbf{1}\{y_i = 0 \wedge \hat{y}_i = 0\},$$

$$\text{FP} = \sum_{i=1}^{N} \mathbf{1}\{y_i = 0 \wedge \hat{y}_i = 1\}, \quad \text{FN} = \sum_{i=1}^{N} \mathbf{1}\{y_i = 1 \wedge \hat{y}_i = 0\}.$$

The confusion matrix is then

$$\mathbf{C} = \begin{pmatrix} \text{TN} & \text{FP} \\ \text{FN} & \text{TP} \end{pmatrix}.$$

For K classes labeled $1, \ldots, K$, let $y_i, \hat{y}_i \in \{1, \ldots, K\}$. The (j, k)-entry of the confusion matrix is

$$C_{jk} = \sum_{i=1}^{N} \mathbf{1}\{y_i = j \wedge \hat{y}_i = k\}, \quad j, k = 1, \ldots, K,$$

so that

$$\mathbf{C} = \left[C_{jk}\right]_{j,k=1}^{K}.$$

3.5 WATER QUALITY PREDICTION

Water quality prediction is a crucial aspect of aquaculture management, as it enables early detection of harmful environmental changes that could affect fish health and growth. By leveraging historical data and real-time sensor inputs, machine learning models can be trained to forecast future water quality parameters such as

3.5.1 pH level

pH measures the acidity or alkalinity of water. Fish have preferred pH ranges, and a deviation from these values can stress them or reduce their growth.Typical Range:Most freshwater fish thrive in a pH range of 6.5 to 7.5, while marine species prefer a pH of around 8.0.Figure 3.3 shows the pH sensor with sensing process.

Figure 3.3: pH sensor

3.5.2 Dissolved Oxygen (DO)

Oxygen is essential for the respiration of aquatic organisms. The amount of dissolved oxygen in water directly affects fish health and growth. Low oxygen levels can result in reduced activity, slower growth, or even mortality. Typical Range: 5-8 mg/L for most species.

3.5.3 Ammonia (NH)

Ammonia is a waste product from fish metabolism and can accumulate in water. In high concentrations, ammonia is toxic to fish and can lead to poor growth or death. Typical Range: Less than 0.25 mg/L is ideal for most species

3.5.4 Nitrites (NO) and Nitrates (NO)

Nitrites are produced during the nitrogen cycle in aquatic ecosystems. High levels can be harmful to fish, while nitrates are generally less toxic but still need to be controlled in aquaculture systems. Typical Range: Nitrites should ideally be at 0 mg/L, while nitrates should be less than 50 mg/L for most species.

3.5.5 Turbidity

Turbidity measures the cloudiness or haziness of water, often caused by particles suspended in it. High turbidity can interfere with fish respiration and feeding, as it may block light and reduce oxygen levels. Typical Range: Clear water with low turbidity is preferred for healthy fish growth.

3.5.6 Hardness (GH and KH)

General hardness (GH) refers to the concentration of dissolved minerals, primarily calcium and magnesium, in water. Carbonate hardness (KH) refers to the buffering capacity of water, helping to maintain stable pH levels. Typical Range: Freshwater fish often do well in moderate hardness (GH of 4-10 dGH), while saltwater fish require higher hardness.

3.6 WEB APPLICATION

The final step in the system is the deployment of a web application, which serves as an accessible interface for users to interact with the fish health prediction model. This application allows users—such as fish farmers, aquaculture technicians, or researchers—to upload water quality data, view predictions about fish health, and receive recommendations or alerts based on the results. Designed to be user-friendly, the web application simplifies the complex process of data analysis and machine learning, making it accessible to individuals without technical expertise.Figure 3.4 shows the Web application Process block diagram.

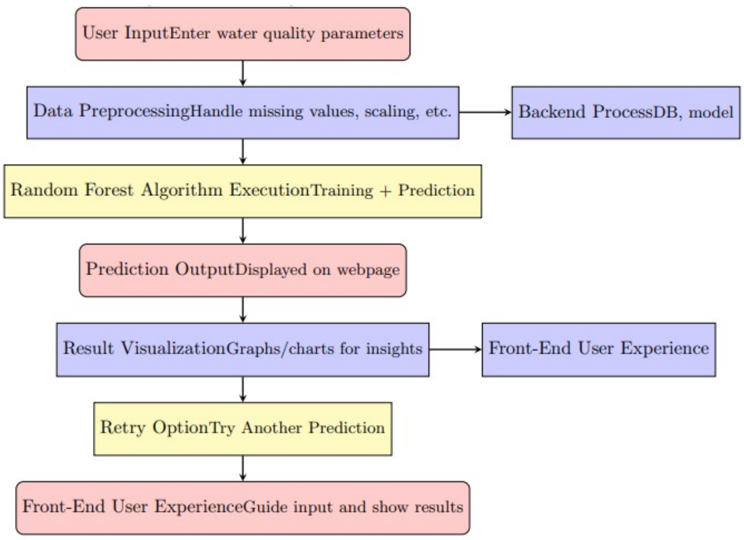

Figure 3.4: Web Process Blockdiagram

1.User Input

The system starts with users entering various water quality parameters into input fields on a web page. These parameters include pH, temperature, turbidity, dissolved oxygen (DO), biochemical oxygen demand (BOD), carbon dioxide (CO), and others. The input interface is designed to be intuitive and clearly labeled. Built-in validation checks ensure the entered values fall within scientifically acceptable ranges, reducing the chance of invalid or harmful input that could lead to incorrect predictions

2.Data Preprocessing

Once the input data is submitted, it goes through a preprocessing pipeline. This involves handling missing values (e.g., using mean or median imputation), ensuring the data format aligns with the model's expectations, and applying feature scaling if necessary. Preprocessing is essential for maintaining the consistency and accuracy of the model's predictions, especially when the input units or scales vary.

3.Backend Process

Behind the scenes, the backend of the application supports all critical operations. It includes a database that may store previous predictions or reference data ranges, a trained Random Forest model, and API or server-side logic that integrates all components. This modular structure allows for easy maintenance, scalability, and future enhancements such as adding new models or logging data for analytics.

4.Random Forest Algorithm Execution

The core of the prediction system is a Random Forest algorithm. This model is trained in advance using labeled datasets that classify water quality as "Safe" or "Unsafe" based on known parameter combinations. When the user submits data, the algorithm evaluates it through an ensemble of decision trees. Each tree provides a classification, and the final prediction is made by aggregating these results through a majority vote mechanism, improving reliability and reducing the risk of overfitting.

5.Prediction Output

After processing the input through the model, the system generates a prediction—typically indicating whether the water is "Safe and Clean" or "Unsafe." This result is immediately displayed on the web interface. The goal here is to offer a quick, comprehensible outcome that is useful for both experts and laypersons concerned with water quality.

6.Result Visualization

To make the outcome more understandable and insightful, the application presents the results with visual aids such as graphs or charts. These illustrations show how different parameters contributed to the final prediction. For instance, users may see that low dissolved oxygen or high turbidity levels were key factors in a negative classification. Visuals not only enhance comprehension but also increase trust in the system's outputs.

7.Front-end User Experience

The user interface is designed with a focus on usability and aesthetics. Incorporating an underwater theme with elements like coral reefs or fish makes the experience engaging and relevant to the subject matter. This visual context helps users relate to the importance of clean water and environmental health while interacting comfortably with the application.

8.Final Front-end Layer

The frontend concludes by reinforcing the overall user experience. It ensures smooth navigation, responsiveness, and accessibility across devices. By guiding users through the entire prediction process clearly and efficiently, it supports both educational and practical use cases for assessing water quality.Figure 3.5 shows the final Web application Front and Bakend layer.

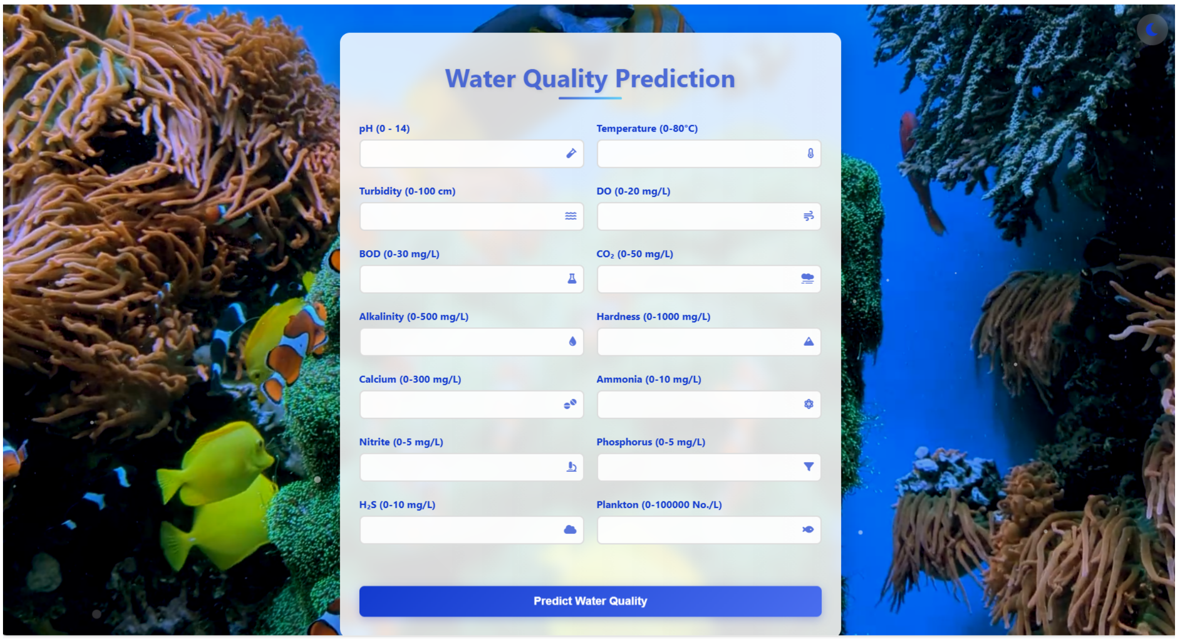

Figure 3.5: Web Application

3.6.1 SUMMARY

This project presents a machine learning-based system for early detection of fish diseases through the analysis of water quality parameters such as pH, dissolved oxygen, ammonia, and turbidity. After preprocessing and selecting key features, various algorithms were tested, with the Random Forest model showing the highest accuracy, identifying pH as a major predictor. A user-friendly web application was developed to allow users to input data, receive instant predictions, and view visual insights. This system supports proactive aquaculture management, promoting fish health and environmental sustainability.

CHAPTER 4

RESULTS AND DISCUSSIONS

4.0.1 ML Algorithm Comparison

Figure 4.1 shows the performance comparison of the algorithms, we can conclude that Decision Tree and Random Forest exhibit the highest accuracy 95 percentage, making them strong candidates for tasks requiring precise classification. SVM also performs well 90 percentage, balancing complexity and effectiveness. Meanwhile, Naive Bayes and KNN show moderate accuracy 85 percentage, which may be suitable for specific datasets but could be outperformed by more sophisticated models. Logistic Regression has the lowest accuracy 80 percentage. The Random Forest model demonstrates high accuracy, making it a reliable choice for predictive tasks. As seen in the model comparison, Random Forest consistently performs well due to its ensemble learning approach, where multiple decision trees work together to reduce overfitting and improve generalization.

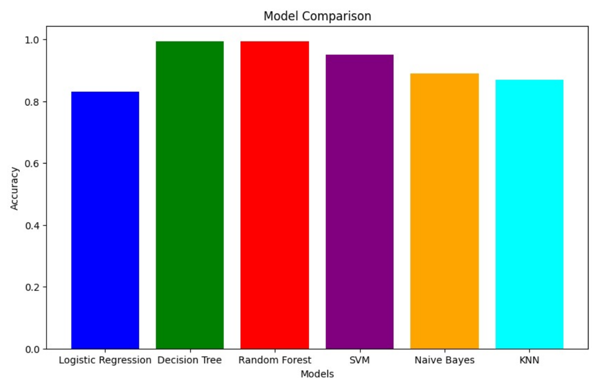

Figure 4.1: ML algorithm comparisation

Case 1: Water is Safe and Clean

From figure4.2 we can applied the bestcase parameters in the web application and get the results as shown in figure4.3.The water quality in this case is excellent, making it safe and clean for various uses. It is suitable for drinking, cooking, and all domestic purposes without the risk of contamination. Aquatic life, including sensitive fish species, can thrive in such water due to optimal

```
Best Case (2) Parameters:
           Temp  Turbidity (cm)  DO(mg/L)  BOD (mg/L)        CO2       pH`  \
1188  28.305607       19.756460  4.089812    4.424872   3.094032  6.710983
109    1.098151       19.216598  6.465732    4.628758   9.999986  6.174835
511   16.451427       28.591500  7.510846    5.300579  14.925399  6.309543
949   19.737057       69.915704  4.172879    4.177671   6.895757  9.232509
1397  34.128671       18.769127  3.603454    5.348025   8.421976  6.879836
...         ...             ...       ...         ...        ...       ...
415   15.863580       29.431197  6.119195   14.541845   2.998492  6.080549
353   28.834295       53.964043  2.335388    1.546088   7.861719  7.240714
1315  30.522598       26.754862  7.084257    1.442935   8.091780  7.809153
1455  34.064720       65.545745  4.943572    1.543270   9.782816  6.162275
1393  32.608606       25.556060  5.577531    1.248952   8.830085  6.700449

      Alkalinity (mg L-1 )  Hardness (mg L-1 )  Calcium (mg L-1 )  \
1188             29.750587          270.408077          78.302718
109             168.734661          245.265025         184.129290
511              29.670747          226.181576          19.720239
949              27.548590          242.567398           9.860786
1397             48.304636          298.559430         154.458052
...                    ...                 ...                ...
415              38.381938           54.555080          15.955020
353              85.181387          137.106133          91.625851
1315             38.273899           87.903041          21.651768
1455             26.273445           44.791112          59.972299
1393            104.480193          186.349303          76.126955
```

Figure 4.2: Bestcase Parameters

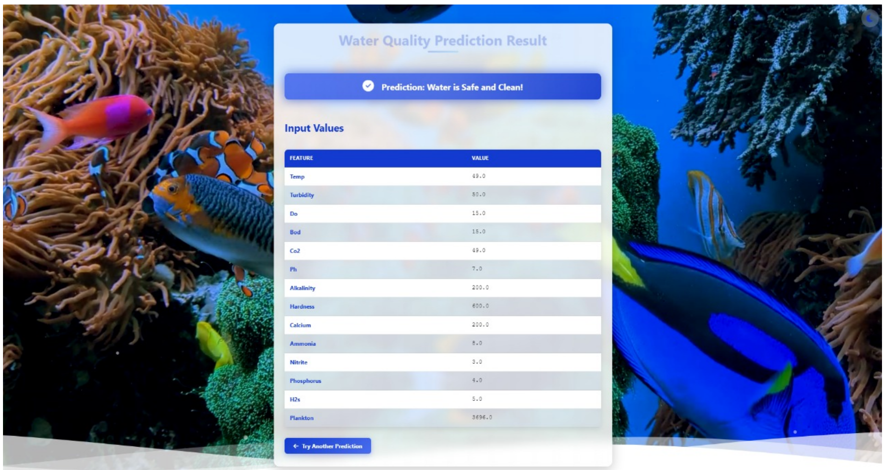

Figure 4.3: Water is Good

levels of dissolved oxygen, stable pH, and low presence of harmful substances. A healthy aquatic ecosystem is supported under these conditions, promoting biodiversity and sustainability. This type of water is also ideal for recreational activities and supports agricultural needs without causing damageing damageor theo crops or the soil. Overall, it represents the best condition for environmental and human health.

Case 2: Water is Moderate

From figure4.4 we can applied the Average case parameters in the web application and get the results as shown in figure4.5.The water quality is moderate, meaning it can be used cautiously for certain non-consumable purposes. While not ideal for drinking, it may be used for irrigation or some industrial processes after minor treatment. Aquatic life is moderately supported—hardy fish species such as tilapia and catfish may survive, but more sensitive species could be negatively impacted.

Parameters like pH, turbidity, and oxygen levels may show slight imbalances, which can lead to long-term ecological concerns if not addressed. Although not heavily polluted, this water is not entirely safe without proper testing and treatment for specific applications.

Average Case (1) Parameters:

	Temp	Turbidity (cm)	DO(mg/L)	BOD (mg/L)	CO2	pH` \
1760	30.338537	15.168593	7.032630	2.479024	4.327260	9.190922
1921	15.477427	26.703443	5.764126	3.725595	1.215205	6.079626
2570	17.678721	20.291242	7.036879	2.571935	1.932373	9.027948
2864	17.300873	22.507733	5.710983	5.370088	3.955638	6.136289
2405	30.898704	26.642223	7.622744	2.190850	9.747657	6.234835
...	...	...	...	...	...	...
1694	17.744556	15.205367	7.678040	4.142980	9.752282	6.191678
2387	17.219490	22.128826	5.734808	2.806380	2.761562	9.464381
1898	18.777245	24.377391	5.273751	3.376220	4.627380	6.327650
2743	15.964328	26.088647	7.218024	3.713003	2.369967	9.297460
2812	15.873862	29.483245	5.814690	3.544321	2.915877	6.125467

	Alkalinity (mg L-1)	Hardness (mg L-1)	Calcium (mg L-1) \
1760	45.626967	204.197177	161.849721
1921	115.442729	271.245686	15.793144
2570	36.578505	248.269481	160.315944
2864	48.732588	151.367737	18.923119
2405	33.814118	47.572262	17.689729
...	...	...	...
1694	196.930117	173.915044	23.803211
2387	41.240696	62.835518	13.342377
1898	33.242004	30.862905	20.582654
2743	40.653565	243.850251	21.904703
2812	159.167004	232.680587	148.701292

Figure 4.4: Averagecase Parameters

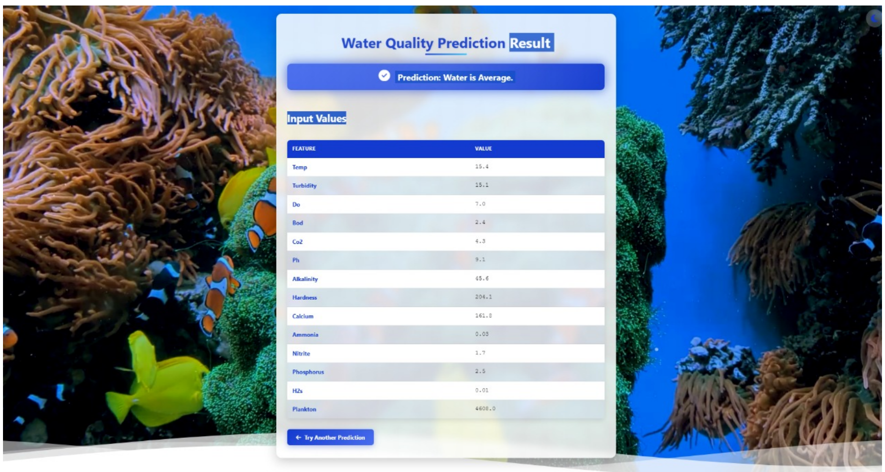

Figure 4.5: Water is Average

Case 3: Water is Poor or Contaminated

From figure4.6 we can applied the Worst case parameters in the web application and get the results as shown in figure4.7.Where the water is considered contaminated and unfit for human consumption or direct contact. Drinking or using this water for domestic tasks poses serious health risks due to the likely presence of harmful pollutants, bacteria, and chemical substances. Aquatic life suffers significantly in this environment, with most fish and other organisms unable to survive because of extremely low oxygen levels, toxic compounds, or harmful pH levels. Such water can damage the surrounding ecosystem, contribute to the spread of disease, and disrupt natural biological cycles.

```
Worst Case (0) Parameters:
           Temp  Turbidity (cm)  DO(mg/L)  BOD (mg/L)       CO2       pH`  \
3035  26.865941       49.634091  4.391976    1.860846  6.688350  6.751112
4041  26.305860       30.423494  4.912412    1.615839  7.348736  7.147743
3771  22.388039       49.441901  4.015676    1.632437  7.848189  8.317536
3241  23.784206       63.526976  4.591504    1.991162  5.824161  6.824204
3250  21.286314       30.020928  4.863378    1.470537  6.188956  8.975399
...         ...             ...       ...         ...       ...       ...
3205  29.130426       48.530501  3.951966    1.014795  6.063397  8.182274
4101  26.785942       63.285066  4.739512    1.542996  6.527943  8.092013
3179  28.929671       73.705072  4.467610    1.546336  7.241073  6.893755
3559  22.391514       63.789627  4.172929    1.732036  7.427659  8.639719
3459  23.040106       66.237537  4.851934    1.851519  5.657814  6.780730

      Alkalinity (mg L-1 )  Hardness (mg L-1 )  Calcium (mg L-1 )  \
```

Figure 4.6: Worstcase Parameters

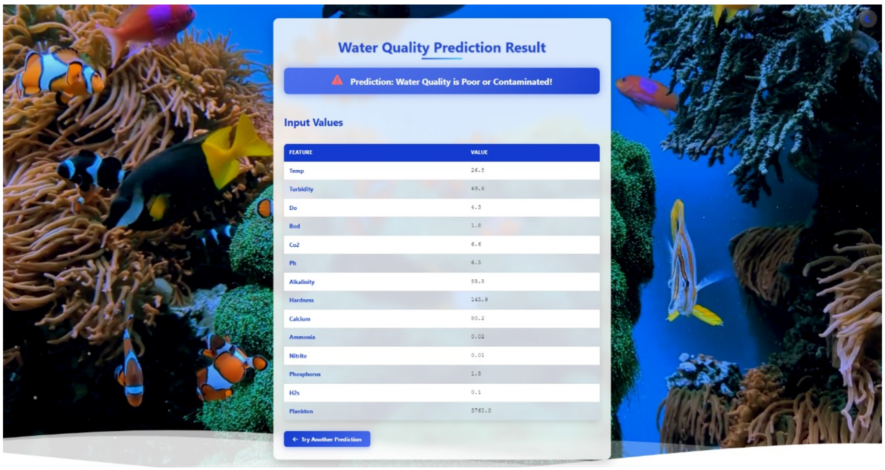

Figure 4.7: Water is Poor and Contaminated

Immediate remediation and restriction of use are strongly recommended.

4.0.2 SUMMARY

The analysis highlights the effectiveness of the Random Forest Classifier in predicting fish diseases based on water quality, with pH emerging as the most influential factor. Three case studies illustrate varying water conditions: Case 1 demonstrates safe and clean water ideal for both human use and aquatic life; Case 2 shows moderate quality water suitable for limited purposes but with ecological risks; and Case 3 reflects contaminated water that poses serious threats to health and aquatic ecosystems. These findings emphasize the importance of monitoring water parameters to maintain ecological balance and prevent disease.

CHAPTER 5

CONCLUSION AND FUTURE

The successful management of aquaculture operations heavily depends on the ability to maintain optimal water quality conditions, as water serves as the living environment for aquatic species. This project aimed to address the challenges of manual and delayed water monitoring by implementing a machine learning-based predictive system that can assess and forecast water quality conditions effectively.

we compared multiple machine learning algorithms and identified Random Forest as the most accurate model for water quality prediction. Its superior performance in classification tasks makes it the ideal choice for ensuring precise and reliable predictions. Based on our findings, we have also developed a web application that effectively predicts water conditions across three distinct cases. This application enables real-time assessment, helping users make informed decisions regarding water usability, environmental impact, and aquaculture management. Our work bridges the gap between machine learning advancements and practical applications, providing a data-driven solution for monitoring and improving water quality.

5.1 FUTURE WORK

Future research in ML-based water quality prediction for aquaculture can be significantly improved by integrating hardware-based monitoring systems. The development of IoT-enabled water sensors will allow real-time measurement of critical parameters such as pH, turbidity, dissolved oxygen, and temperature, ensuring more accurate predictions. Additionally, incorporating edge computing with microcontrollers like Raspberry Pi or ESP32 can optimize processing speed, reducing reliance on cloud computing and enabling on-site analysis. AI-driven adaptive models will allow dynamic adjustments in response to environmental changes, leading to better predictive accuracy. Moreover, integrating a decision-support system that regulates water conditions automatically can enhance sustainability and fish health. These advancements will contribute to an intelligent, real-time monitoring system, enabling efficient, sustainable, and high-yield aquaculture management.

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